

1. Use the Windows command prompt to run the command `ipconfig /all` on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

```
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : bbrouter
Description . . . . . : Broadcom 802.11n Network Adapter
Physical Address. . . . . : 78-E4-00-12-AD-7C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2001:db8:1000:1::10(Preferred)
IPv6 Address. . . . . : 2402:a00:408:5605:b47b:d78:cf2:784d(Preferred)
Temporary IPv6 Address. . . . . : 2402:a00:408:5605:1c7d:4da4:3d04:bb83(Preferred)
Link-local IPv6 Address . . . . . : fe80::6962:fd55:6f5d:1781%15(Preferred)
IPv4 Address. . . . . : 192.168.1.2(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Tuesday, August 11, 2026 6:14:39 AM
Lease Expires . . . . . : Wednesday, August 12, 2026 6:31:58 AM
Default Gateway . . . . . : 2001:db8:1000:1::1
                          fe80::6231:92ff:fe04:38bd%15
                          192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 158917632
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-9F-17-4E-00-26-B9-DE-88-FB
DNS Servers . . . . . : fe80::6231:92ff:fe04:38bd%15
                          192.168.1.1
NetBIOS over Tcpip. . . . . : Enabled
```

2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to `www.google.com`. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

Network Connectivity Simulation

You can perform this safely on your own computer by disconnecting Wi-Fi or unplugging the Ethernet cable.

1. Disconnect Wi-Fi/Ethernet and run:

- `ping www.google.com`

Expected result:

- Ping request could not find host `www.google.com`.

or

- General failure.

Meaning: The computer has no working network connection, so it cannot reach Google or may not be able to resolve the domain name.

2. Reconnect Wi-Fi/Ethernet and run again:

- ping www.google.com

Expected result:

- Reply from <IP address>: bytes=32 time=20ms TTL=...
- Reply from <IP address>: bytes=32 time=18ms TTL=...

Meaning: The computer has Internet connectivity and can reach Google's server. time shows the approximate round-trip delay.

3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time

34% CPU	78% Memory	19% Disk	1% Network
23.2%	930.0 MB	0.1 MB/s	1.9 Mbps
0%	39.2 MB	0 MB/s	0 Mbps
2.3%	21.7 MB	0.1 MB/s	0 Mbps

4. Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.

Wireshark Packet Capture – Flipkart/Instagram Web

You can complete this on your own laptop using Wireshark.

Steps:

1. Open **Wireshark**.
2. Select your active **Wi-Fi or Ethernet** interface.
3. Click **Start Capture**.
4. Open Flipkart or Instagram Web in your browser and browse for about **30 seconds**.
5. Return to Wireshark and click **Stop**.
6. In the display filter box, enter:
 - http

For HTTPS/TLS traffic, use:

- tls
7. Press **Enter** to display the filtered packets.

Information	What You Can Observe
1. Source & Destination IP addresses	The IP address of your computer and the remote server involved in the connection.
2. Protocol and Port Information	Protocols such as TCP/TLS and ports such as 443 used for HTTPS.
3. Packet Timing & Size	The time a packet was captured, packet length, and sequence/acknowledgment information.

5. Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and

reliability. Paste the AI's response and highlight one suggestion you would try first.

AI-Generated Troubleshooting Checklist

1. **Restart the router and modem** – Power them off for 30 seconds and turn them back on to refresh the connection.
2. **Use Ethernet instead of Wi-Fi** – A wired connection usually provides lower latency and greater stability for gaming.
3. **Move closer to the Wi-Fi router** – Reduce distance and obstacles between the gaming device and router.
4. **Stop background downloads** – Pause streaming, cloud backups, Windows updates, and other activities using bandwidth.
5. **Check for other connected devices** – Disconnect unnecessary devices that may be consuming network bandwidth.
6. **Update router firmware and network drivers** – Updates can improve performance, stability, and security.
7. **Choose the less congested Wi-Fi band** – Use **5 GHz or 6 GHz** when available and suitable for the device.